ANNEXURE A – DRY RUN EVALUATION

1. Question 1 – Breadth-First Search (BFS)

Given graph (left child is visited before right child):

A\n / \\\n B C\n / \\ / \\\n D E F G

BFS visits nodes level by level, from left to right.

Iteration	Queue Before Processing	Visited Node	Queue After Processing
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

1. BFS traversal order: A → B → C → D → E → F → G

2. Queue contents after each iteration: (1) B, C; (2) C, D, E; (3) D, E, F, G; (4) E, F, G; (5) F, G; (6) G; (7) Empty.

2. Question 2 – Depth-First Search (DFS)

DFS explores one branch as deeply as possible before backtracking. The left child is visited first.

Iteration	Visited Node	Active Path (Stack Concept)
1	A	A
2	B	A, B
3	D	A, B, D
4	E	A, B, E
5	C	A, C
6	F	A, C, F
7	G	A, C, G

DFS traversal order: A → B → D → E → C → F → G

Note: The stack column shows the active path conceptually; an iterative DFS stack may differ depending on when children are pushed and popped.

3. Question 3 – 8-Puzzle Using Breadth-First Search (BFS)

Initial State:

1 3 6\n5 0 2\n4 7 8

Goal State:

1 2 3\n4 5 6\n7 8 0

Here, 0 represents the empty space. For compact state notation, "_" represents the empty space and "/" separates rows.

A. BFS dry-run table (first five iterations)

Iteration	Current State (Expanded)	Queue Before Expansion	Queue After Expansion
1	136/5_2/478	136/5_2/478	1_6/532/478; 136/_52/478; 136/52_/478; 136/572/4_8
2	1_6/532/478	1_6/532/478; 136/_52/478; 136/52_/478; 136/572/4_8	136/_52/478; 136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478
3	136/_52/478	136/_52/478; 136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478	136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78
4	136/52_/478	136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78	136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78; 13_/526/478; 136/528/47_
5	136/572/4_8	136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78; 13_/526/478; 136/528/47_	_16/532/478; 16_/532/478; _36/152/478; 136/452/_78; 13_/526/478; 136/528/47_; 136/572/_48; 136/572/48_

B. Questions and Answers

1. Which node is expanded in each iteration?

The first five expanded states are:

- Iteration 1: 136/5_2/478
- Iteration 2: 1_6/532/478
- Iteration 3: 136/_52/478
- Iteration 4: 136/52_/478

- Iteration 5: 136/572/4_8

2. How many states are generated in total?

Using BFS with duplicate-state checking and the move order used in this trace:

- Distinct states discovered by the time the goal is reached: 456 (including the initial and goal states).
- States removed from the queue up to and including the goal: 283.
- States in the solution path: 9 (including the initial and goal states).

The total can differ depending on the definition of "generated" and the implementation.

3. Complete solution path from the initial state to the goal state:

Initial	1 3 6\n5 0 2\n4 7 8
Move 1	1 3 6\n5 2 0\n4 7 8
Move 2	1 3 0\n5 2 6\n4 7 8
Move 3	1 0 3\n5 2 6\n4 7 8
Move 4	1 2 3\n5 0 6\n4 7 8
Move 5	1 2 3\n0 5 6\n4 7 8
Move 6	1 2 3\n4 5 6\n0 7 8
Move 7	1 2 3\n4 5 6\n7 0 8
Goal – Move 8	1 2 3\n4 5 6\n7 8 0

The solution path requires 8 moves.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

BFS explores states level by level. It first explores states requiring one move, then two moves, then three moves, and so on. Since every move has equal cost, the first time BFS reaches the goal, the solution has the minimum number of moves.